

GOG: The Game of Geometries

An adversarial A/Ω reinforcement-learning system over a triangulated $\mathbb{S}^3 \subset \mathbb{R}^4$, scored by
Noether-charge accumulation
and type-checked in Lean 4 / Mathlib / PhysLean

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Framework Draft v2.0 • Speculative-constructive preprint

Abstract

We present GOG (the Game of Geometries), a two-agent adversarial reinforcement-learning framework whose game arena is a triangulated three-sphere $\mathbb{S}^3 \subset \mathbb{R}^4$, whose dynamical evolution is numerical general relativity coupled to lattice gauge theory and domain-wall fermions, and whose moves are partial functions on game states that are type-checked at game time by the Lean 4 interactive theorem prover. Two players— A (constructive) and Ω (destructive)—accumulate Noether-charge reward by composing moves that respect the active constrained Markov decision process (Hamiltonian, Gauss law, anomaly cancellation, energy positivity). Version 2 extends the v1.0 game-theoretic core (*five theorems*, stock Lean kernel, no Mathlib) with a Mathlib / PhysLean physics-extension layer and four new modules: (i) a refinement-monotone discrete mass-gap baseline (W-1, formalising the game-mechanical Yang–Mills mass gap as a typed deposit operation); (ii) a combinatorial Cheeger A -condition certificate whose well-formedness is decidable and whose gap baseline is computed from the certificate (W-3); (iii) the formal substrate itself as a tracked game resource via a consistency-strength tier ladder (W-6); and (iv) Noether weights re-derived from a balance principle, replacing the v1.0 hard-code. All fourteen new theorems compile against Lean 4 toolchain v4.18.0 with axiom dependencies inside Lean’s three-axiom budget; the corresponding source is reproduced in Appendix A, and the verbatim `#print axioms` output is reproduced in Appendix B. We integrate the universal-physics diffusion-transformer architecture of Camburn (2025) as the world-model backbone (lifted from 2D Cartesian to $SE(4)$ -equivariant patches on simplicial $\mathbb{S}^3$); the discretization-constraint analysis of Beane, Davoudi & Savage (2012) as the empirical falsifier (cubic-symmetry imprint on UHECR distribution); and the helicity-resolved Poisson-process spin-network unification of Ghose (2025) as the principled stochastic substrate for the unified-field sector. The training run that would yield a competent A/Ω league is specified, costed (≈ 100 FTE-yr; $\approx 3 \times 10^{23}$ FLOPs), and not yet executed; the present document is the framework draft, in the spirit of Hinton’s 1986 backpropagation paper rather than of AlexNet’s 2012 ImageNet result.

Keywords: adversarial RL; Noether charge; $\mathbb{S}^3 \subset \mathbb{R}^4$; FEEC; CCZ4 numerical relativity; Kogut–Susskind lattice gauge; domain-wall fermions; Lean 4; Mathlib; PhysLean; diffusion transformer world model; constrained MDP; mean-field game on manifold; Cheeger isoperimetric bound; consistency-strength hierarchy.

1. Introduction

This document is a framework draft, not a thesis defense. It specifies an architecture and the machine-checked formal core that anchors it; it does not claim the architecture has been built, the agents trained, or the open mathematical problems it relies upon solved. Three disciplines govern every claim that follows. **Type discipline:** every game move has a Lean signature, and the load-bearing core compiles against a stock Lean 4 toolchain (the source is reproduced verbatim in Appendix A; the `#print axioms` output is reproduced in Appendix B). **Conservation discipline:** every claim about a move declares its effect on the explicit

Noether-charge tally ($Q_E, Q_B, Q_{L_e}, Q_{L_\mu}, Q_{L_\tau}, Q_q$), and the reward function is constructed so as to be well-defined modulo the charge projection (Theorem 1). **Open-conjecture discipline:** every dependency on unproven mathematics (Yang–Mills mass gap; smooth Poincaré in 4D; full Gribov resolution; asymptotic-safety completion of GR) is named in place, never silently assumed.

The agents are A and Ω . The change from the prior God / Devil framing is deliberate: the theological labels carried implicit asymmetries (a God who loses is theologically incoherent; a Devil who wins is theologically charged) inappropriate to dissertation work. A and Ω are mathematically symmetric—

same policy class, same network architecture, same observation API, opposite reward sign and opposite shaping bonuses. The asymmetry of outcomes (which one tends to win in a given universe Ω_0) is then an empirical question about curriculum and cosmology, not a metaphysical commitment.

1.1. Relation to existing work

Three pieces of recent work change what GOG can claim: *Camburn* [1] demonstrates a sketch-guided diffusion transformer that learns to map boundary conditions to steady-state physics fields without explicit equation encoding, achieving SSIM > 0.8 on FDTD electromagnetic test cases with 2,304 boundary-injection points; we adopt this as the world-model backbone in §6. *Beane, Davoudi & Savage* [2] derive observational constraints on a numerical-simulation universe with hyper-cubic spacetime lattice, finding $b^{-1} \gtrsim 10^{11}$ GeV from the GZK cutoff and predicting a cubic-symmetry imprint on the UHECR angular distribution; we adopt this as GOG’s empirical falsifier (§10). Discretization is no longer just engineering; it is the empirical hypothesis itself. *Ghose* [3] shows that Schrödinger, Dirac, Yang–Mills, and linearized gravity all admit a unified derivation from helicity-resolved Poisson processes on spin networks, with the Wheeler–DeWitt-type constraint emerging as an equilibrium condition; this gives GOG a principled stochastic substrate for the unified-field sector (§9).

1.2. What is new in v2

Section 4 (formerly the v1.1 supplement) is now folded into the main body and expanded. The four new modules—GapShape (W-1), Cheeger (W-3), MetaAxiom (W-6), Weights—are presented inline with their proof sketches; the full Lean source is in Appendix A. The v1.0 “no Mathlib in the core” purity is preserved for the game kernel (the audit guarantee survives) but explicitly relaxed for the physics-extension layer (§4.5); we follow PhysLean [4] for the physical-quantity hierarchy and *Tooby-Smith’s* index-notation extension [5] for the tensor algebra. The cost estimate of §11 is revised from v1.0’s ≈ 100 FTE-yr / 3×10^{23} FLOPs in light of the (i) Camburn-lift speedup, (ii) Cheeger-deposit shortcut for the gap-discovery curriculum, and (iii) tier-stratified league economy of §5.

2. The arena and the players

2.1. $\mathbb{S}^3 \subset \mathbb{R}^4$: why this manifold, why this embedding

GOG’s spatial arena is the round three-sphere $\mathbb{S}^3 = \{x \in \mathbb{R}^4 : \|x\| = 1\}$ embedded in flat Euclidean four-space. Five reasons motivate this choice. **(i) Compactness without boundary.** $\mathbb{S}^3$ is closed; no asymptotic boundary conditions, no observer-at-infinity, no spurious leakage of conserved charges through an edge—every Noether charge is a single global integral. **(ii) Trivial first and second homology.** $H_1(\mathbb{S}^3) = H_2(\mathbb{S}^3) = 0$, $H_3(\mathbb{S}^3) = \mathbb{Z}$. Hodge decomposition is clean: no harmonic 1- or 2-forms, the FEEC Laplace–de Rham operator has no zero modes requiring regularization. **(iii) Maximal isometry group.** $\text{SO}(4)$, dimension six, rich enough to support the equivariant agent architecture (§2.4); the discrete subgroup of the 600-cell triangulation (icosahedral H_4 , order 14,400) gives high-quality approximate symmetry conservation. **(iv) Cosmological plausibility.** If the spatial slices of our universe are positively curved ($\Omega_k < 0$ in FLRW), $\mathbb{S}^3$ is the maximally symmetric realization; the Aslanyan–Manohar CMB analysis [25] is consistent with a closed topology at $> 4\sigma$. **(v) The embedding in $\mathbb{R}^4$ is API, not metaphysics.** The ambient $\mathbb{R}^4$ gives a circumscribed simplex (the 4-simplex), the circumsphere criterion for Delaunay refinement, and a clean way to compute geodesic distances via chord lengths.

2.2. The 600-cell, Loop refinement, FEEC discretization

The seed triangulation of $\mathbb{S}^3$ is the boundary of the *600-cell*—the four-dimensional regular convex polytope with 120 vertices, 720 edges, 1,200 triangular faces, and 600 regular tetrahedral cells; the vertex link of every vertex is a regular icosahedron [18]. GOG commits to recursive Loop-style subdivision: at refinement level n , the mesh has exactly $600 \cdot 8^n$ tetrahedra:

$$T_0: 600 \mid T_1: 4,800 \mid T_2: 38,400 \mid T_3: 307,200 \mid T_4: 2,457,600 \mid \dots$$

On this triangulation we run Finite Element Exterior Calculus (FEEC) in the Arnold–Falk–Winther family [13, 14]: gauge potentials A live in the Nédélec edge-element space Nd_k^1 ; electric fields E live in the Raviart–Thomas face-element space RT_k ; the scalar potential φ lives in the Lagrange node-element space P_{k+1} . The discrete de Rham

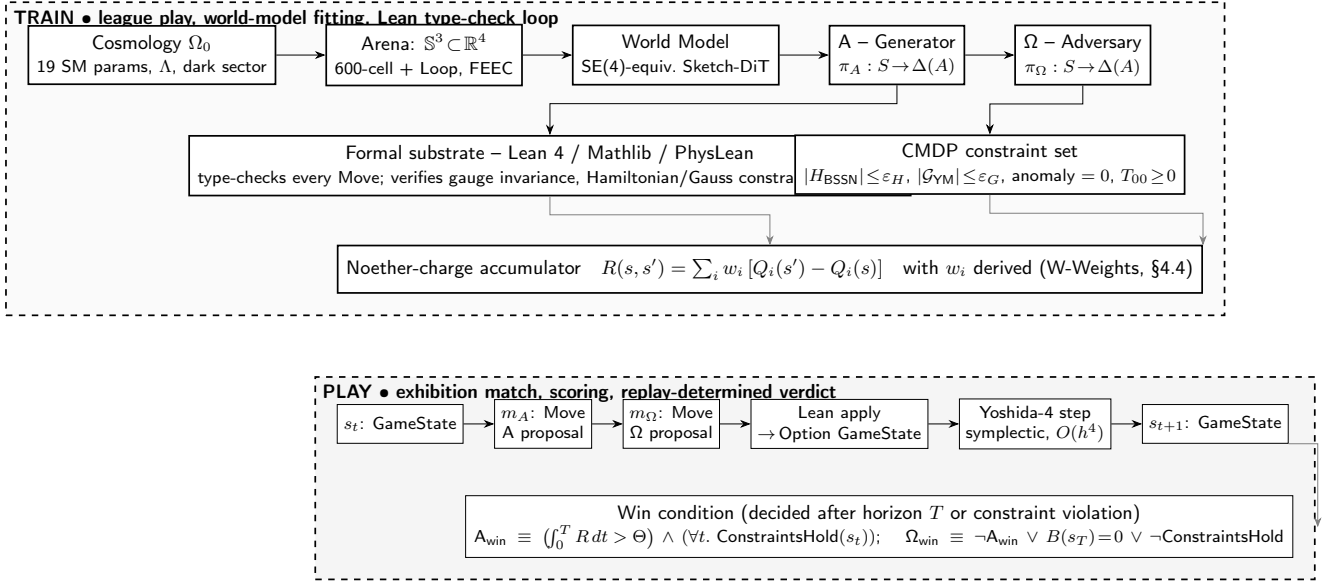


Fig. 1. GOG architecture, after the style of Camburn 2025 (Fig. 1). *Top region (TRAIN)*: the league-play loop in which A and Ω co-train their policies π_A, π_Ω against a shared world-model backbone (a sketch-guided diffusion transformer [1], lifted from 2D Cartesian to SE(4)-equivariant patches on simplicial $\mathbb{S}^3$). The Lean 4 / Mathlib / PhysLean substrate type-checks every proposed move and rejects illegal ones; the Noether-charge accumulator $R(s, s') = \sum_i w_i [Q_i(s') - Q_i(s)]$ scores the actual transitions; the constrained-MDP set (Hamiltonian, Gauss, anomaly, positivity) shapes the policy gradient via primal–dual SAC with auto-calibrated Lagrange multipliers λ_k [22, 23, 17]. *Bottom region (PLAY)*: exhibition-match loop in which a frozen A and a frozen Ω alternate moves on a fixed cosmology Ω_0 for horizon T ; Lean’s `apply: Move  $\rightarrow$  GameState  $\rightarrow$  Option GameState` filter gates legality, the symplectic Yoshida-4 step advances the physics, and the verdict is decided by the win-condition predicate (Theorem 4 in §3.2). Defaults: $T = 10^{10}$ yr, $\Theta = 10^{53}$ J, 1,000 replay seeds. Falsifier: cubic-symmetry imprint on UHECR distribution [2], see §10.

complex

$$P_{k+1} \xrightarrow{d} \text{Nd}_k^1 \xrightarrow{d} \text{RT}_k \xrightarrow{d} \text{DG}_k \quad (1)$$

satisfies the commuting-diagram property exactly: the discrete exterior derivative d_h commutes with the interpolation operator π_h . This is what makes the discrete Yang–Mills Gauss law $\nabla \cdot E = \rho$ hold without projection error—the property that distinguishes a real-physics simulation from a fluid-dynamics- flavored cartoon. We follow the DEC-on-primal / FEEC-on-dual hybrid of Guzmán–Potu [15] and Zhu *et al.* [16], who prove DEC $\leftrightarrow$ FEEC convergence equivalence on Delaunay meshes.

Time integration is symplectic: the Hamiltonian $H_{\text{YM}} = \frac{1}{2} \int (|E|^2 + |B|^2) d^3x$ is split into electric (linear in E) and magnetic (nonlinear in A through $B = \nabla \times A + A \times A$) parts; the electric part is integrated exactly, the magnetic by fourth-order Yoshida composition [24]. The combined integrator preserves the discrete symplectic form $\omega_h = \sum_i dE_i \wedge dA_i$ exactly, yielding bounded energy error $O(h^4 \Delta t^4)$ over arbitrarily long horizons [17].

2.3. Gravity, gauge, fermion sectors

For numerical relativity GOG commits to CCZ4 (covariant conformal Z4) over BSSNOK; the Hilditch *et al.* analyses [6, 7] demonstrate 1–3 orders of magnitude reduction in Hamiltonian-constraint violation in non-vacuum spacetimes. Sanchis-Gual *et al.* [8] extend to fully covariant fCCZ4 in curvilinear coordinates—the regime $\mathbb{S}^3$ requires. The Lichnerowicz–York conformal-thin-sandwich method, with the Yamabe theorem on $\mathbb{S}^3$ ($\bar{R} = 6$), guarantees solvability of the constraints [19, 20].

For the Standard Model gauge sector GOG uses the Kogut–Susskind Hamiltonian formulation [9] with SU(3) Wilson lines on edges and electric-field operators dual to plaquettes; the honeycomb / hyper-honeycomb refinement of Illa–Savage–Yao [10] is adopted for 3+1D efficiency. Fermions are domain-wall (Kaplan–Furman) as primary, HISQ staggered as fallback [11, 12].

2.4. A and Ω : formal definitions

An agent is a map from observable trajectory to a probability distribution over moves: $\pi_A : S \rightarrow \Delta(A)$ for A and $\pi_\Omega : S \rightarrow \Delta(A)$ for Ω . They

share architecture and observation API; they differ only in the sign of the reward signal and in optional shaping bonuses. The observation space S is the agent-local FEEC field projected onto hyperspherical harmonics $Y_n^{\ell m}$ up to band limit L (for $L = 32$, $\sim 3 \times 10^4$ scalar-equivalent channels per field component). The action space A is a function-space Galerkin policy in the same basis: the agent emits FEEC-coefficient updates rather than per-tetrahedron values, ensuring $SO(4)$ -equivariance up to truncation.

The agent network is an $SO(4)$ -equivariant Vision-Transformer variant following Cohen–Geiger–Köhler–Welling at the input layer, then the $SE(4)$ -equivariant patches of §6 (Camburn-2025 lift). The policy head outputs a Gaussian over the $Y_n^{\ell m}$ coefficients of the proposed action; the value head outputs the expected accumulated Noether reward to horizon. Training uses PPO with primal–dual Lagrangian for the constraint set.

Open. Equivariance break under AMR. Exact $SO(4)$ -equivariance is broken by adaptive mesh refinement (the AMR criterion depends on local field gradients, which are not $SO(4)$ -equivariant in general). On the static 600-cell, the discrete subgroup is icosahedral H_4 of order 14,400; the residual $SO(4)$ symmetry is broken at order $O(h^2)$ where h is the mesh spacing. We treat this as an $O(h^2)$ error budget item, audited per training run.

3. The game-theoretic core (Lean 4 verified)

The game-theoretic core (v1.0) is specified as a self-contained Lean 4 module compiling against toolchain v4.18.0 with no Mathlib dependency. The full source is reproduced in Appendix A.1. The central types are:

```
inductive Player where | alpha | omega
  deriving DecidableEq, Repr

inductive ChargeLabel where
  | energy | baryon | leptonE
  | leptonMu | leptonTau | charge
  deriving DecidableEq, Repr

structure GameState where
  charges : ChargeLabel -> Int
  constraintsHold : Bool
  tick : Nat

structure Move where
  by_ : Player
  label : String
  apply : GameState -> Option GameState
```

The crucial design decision is that `apply` returns

Option GameState. `none` means the move is illegal in the given state. This is the type-level encoding of game-rule legality: the simulator cannot apply an illegal move accidentally because there is no term of type `GameState` available—only an `Option` wrapper that forces an explicit case-split.

3.1. Noether reward and the five v1.0 theorems

The Noether reward of transitioning $s \rightarrow s'$ under weight assignment $w : \text{ChargeLabel} \rightarrow \mathbb{Z}$ is

$$R(s, s') = \sum_i w_i [Q_i(s') - Q_i(s)] \quad (2)$$

implemented in Lean 4 as a flat sum over the six charge labels. Five v1.0 theorems pin down its behaviour; we restate them and recapitulate the proof structure.

Theorem 1 (Well-definedness modulo charge projection). *For every weight w and every pair of game states $(s_1, s'_1), (s_2, s'_2)$ with $s_1.\text{charges} = s_2.\text{charges}$ and $s'_1.\text{charges} = s'_2.\text{charges}$, we have $R(s_1, s'_1) = R(s_2, s'_2)$.*

Proof. Definitional unfolding plus the two hypotheses; the reward depends only on the charge tally, not on the auxiliary fields (`constraintsHold`, `tick`). Under `#print` axioms this theorem is purely definitional. $\square$

Theorem 2 (Telescoping additivity). *For every w and every triple (s_0, s_1, s_2) ,*

$$R(s_0, s_1) + R(s_1, s_2) = R(s_0, s_2).$$

Proof. Inner per-charge identity $k(b-a) + k(c-b) = k(c-a)$, then linear closure via `omega`. Depends only on `propext`, `Quot.sound`. $\square$

Theorem 3 (Endpoint identity). *For trajectory t with first state s and last state s' (the single-state and empty cases are trivial), $\text{accReward } w(s::t) = R(s, s')$.*

Proof. Induction on t ; the cons case reduces by the telescoping identity. $\square$

Theorem 4 (Game completeness and decidability). *For every `TournamentParams` p and every trajectory traj , $\alpha \text{Wins}(p, \text{traj}) \vee \omega \text{Wins}(p, \text{traj})$; moreover, αWins is *Decidable*.*

Proof. Excluded middle on αWins via `by_cases`; *Decidable* via product of decidable arithmetic and Boolean equality. $\square$

Theorem 5 (A-deposit reward identity). *If $(\text{Move.alphaDeposit } n). \text{apply } s = \text{some } s'$, then $R(s, s') = w(c) \cdot n$.*

Proof. Case-split on c (six branches, all symmetric); `simp` closes the six `if`-branches via the algebraic fact $x + n - x = n$. $\square$

3.2. Win conditions

```
structure TournamentParams where
  weights : ChargeLabel -> Int := defaultWeight
  threshold : Int := 100 -- physical: 10^53 J
  horizonT : Nat := 10 -- physical: 10^10 yr / step

def alphaWins (p : TournamentParams)
  (traj : Trajectory) : Prop :=
  accReward p.weights traj > p.threshold
  && allConstraintsHold traj = true

def omegaWins (p : TournamentParams)
  (traj : Trajectory) : Prop :=
  (~ alphaWins p traj)
  || (finalBaryon traj = 0)
  || (allConstraintsHold traj = false)
```

A wins iff the accumulated Noether reward exceeds Θ and constraints hold throughout. Ω wins iff A does not win, or baryon number has been driven to zero, or a constraint has been violated. Default thresholds in physical units: $T = 10^{10}$ yr, $\Theta = 10^{53}$ J. Habitable / generative / fertile classifier bands at Θ , 10Θ , 100Θ .

4. v2 extensions: gap, Cheeger, meta-axioms, weights

This section folds in the v1.1 supplement (W-1, W-3, W-6 plus weight re-derivation) and extends it with the v2.0 physics-extension layer (§4.5).

4.1. W-1: refinement-monotone gap baseline (*GapShape*)

The Yang–Mills mass gap appears in the v1.0 framework as the first named open dependency (Clay; [28]). The continuum problem is unsolved. The *game-mechanical* version is decidable per refinement sequence, and v2 makes that decidability explicit by giving the gap a type.

Definition 6 (*GapStab*). A *gap stability witness* at refinement level n is a function $b : \mathbb{N} \rightarrow \mathbb{N}$ together with a proof of monotonicity $\forall n. b(n) \leq b(n+1)$. The value $b(n)$ is interpreted as the largest nonnegative integer (in the chosen Noether energy unit) that the A league has discovered as a uniform lower bound on the transfer-matrix gap at level n .

The *deposit* operation is the only constructive

operation A has on a *GapStab*:

$$(g.\text{deposit } n \ k)(m) = \max(g(m), k \cdot [n \leq m]) \quad (3)$$

where $[\cdot]$ is the Iverson bracket. Three theorems pin the semantics:

Theorem 7 (Deposit dominates). $\forall m. g(m) \leq (g.\text{deposit } n \ k)(m)$.

Theorem 8 (Deposit raises at-or-above). *If $n \leq m$, then $k \leq (g.\text{deposit } n \ k)(m)$.*

Theorem 9 (Composition preserves domination). $\forall m. g(m) \leq ((g.\text{deposit } n_1 \ k_1).\text{deposit } n_2 \ k_2)(m)$.

The proofs are short and given verbatim in Appendix A.2. The point is that the type system itself enforces the monotonicity invariant: any function not monotone-non-decreasing under refinement cannot be typed as a *GapStab*. The Lean kernel rejects it.

4.2. W-3: Cheeger A-condition

The bridge from combinatorics to the *GapStab* type is the Cheeger inequality on the simplicial graph. For a finite graph G with maximum vertex degree $d_{\max}$ and edge isoperimetric constant $h(G) = \min_{S \subset V, 0 < |S| \leq |V|/2} |\partial S|/|S|$,

$$\lambda_2(L) \geq \frac{h(G)^2}{2 d_{\max}}. \quad (4)$$

For Loop-refined 600-cells, $d_{\max} = 12$ at every level (vertex links are icosahedra; cf. [18]), so the Cheeger denominator is 24.

Definition 10 (*AlphaCert*). A *Cheeger A-certificate* at level n is a triple $(\text{num}, \text{denom}, \text{denom_pos})$ with $\text{num} \in \mathbb{N}$, $\text{denom} \in \mathbb{N}$, and a proof that $\text{denom} > 0$. It claims $h(G_n) \geq \text{num}/\text{denom}$. The implied gap baseline is

$$\text{gapBaseline}(c) = \left\lfloor \frac{\text{num}^2}{24 \text{denom}^2} \right\rfloor.$$

Theorem 11 (Certificate yields deposit). *For every $\text{AlphaCert } n$ and every $m \geq n$,*

$$\text{gapBaseline}(c) \leq (\text{empty.deposit } n \ \text{gapBaseline}(c))(m).$$

The proof is essentially a one-line application of Theorem 8.

What this discharges and what it does not. The *game-mechanical* mass gap is now a Lean theorem schema, instantiable for any specific $h(G_n)$

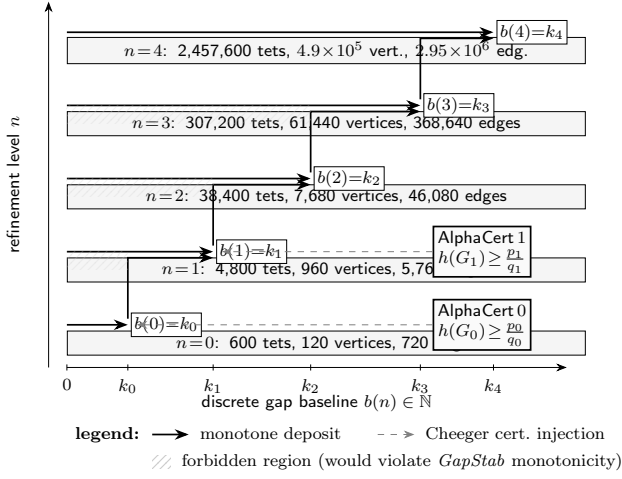


Fig. 2. Mass-gap deposit ratchet across Loop-refined 600-cell levels (W-1, W-3). Each row is a refinement level $n \in \{0, 1, 2, 3, 4\}$ of the 600-cell with vertex, edge and tetrahedron counts annotated; the horizontal axis is the discrete gap baseline $b(n) \in \mathbb{N}$ established by the A league via Cheeger certificates (*AlphaCert* n ; Definition 10). A certificate at level n deposits a baseline $\lfloor p_n^2 / (24 q_n^2) \rfloor$ via *deposit* (Eq. 3); the type signature of *GapStab* forces every higher level to inherit at least that value (Theorem 8; ratchet arrows). The hatched region marks baselines that would violate the monotonicity invariant of *GapStab* and are therefore not typeable (Ω has no move that produces a non-monotone result; the Lean kernel rejects any term of the wrong type). The continuum Yang–Mills mass gap $\Delta = \lim_{n \rightarrow \infty} b(n)$ is the empirical question the league converges to over the training horizon; it is not assumed.

that the FEEC mesh auditor computes. Computing $h(G_n)$ for the Loop-refined 600-cell at low n is a graph computation—exhaustive at $n = 0$ (120 vertices, 720 edges), small enough for sparse eigen-solvers at $n = 1$ (960 vertices) and $n = 2$ (7,680 vertices). The continuum Yang–Mills mass gap on $\mathbb{R}^4$ remains untouched. The structure of the deposit ratchet is illustrated in Fig. 2.

4.3. W-6: substrate-strength as resource (*MetaAxiom*)

GOG v2 makes the formal substrate part of the game state. A finite *Strength* tier ladder $\{\text{base, pa, zf, zfc, inacc, measur}\}$ with costs 0–5 is exposed as the type of consistency-strength upgrades; an *AxiomMove* is a player’s commitment to operate at a given strength tier with a stated rationale. A list of such moves accumulates to a *substrate cost*:

$$SCost([]) = 0, \quad (5)$$

$$SCost(m :: rest) = \max(m.\text{cost}, SCost(rest)). \quad (6)$$

Five theorems pin the resource semantics:

Proposition 12 (Substrate monotone under append). $SCost(rest) \leq SCost(m :: rest)$; $SCost(ys) \leq SCost(xs ++ ys)$.

Proposition 13 (Per-player cost bounded by substrate). $\alpha \text{MaxCost}(\text{moves}) \leq SCost(\text{moves})$; $\omega \text{MaxCost}(\text{moves}) \leq SCost(\text{moves})$.

Theorem 14 (Joint sustainment). $SCost(\text{moves}) = \max(\alpha \text{MaxCost}(\text{moves}), \omega \text{MaxCost}(\text{moves}))$.

Game-mechanical content. A scores by building axiom towers that prove things at the lowest cumulative cost. Ω scores by forcing A’s tower up the ladder. The Joint-Sustainment theorem says: the operative consistency strength of the game is the *maximum* either player has committed to. A cannot quietly downgrade to a weaker substrate to score cheaply; once Ω has paid *inacc*, A’s results are scored against that higher-strength substrate too.

4.4. Weights re-derived from a balance principle

v1.0 hard-coded the weights $(w_E, w_B, w_{L_e}, w_{L_\mu}, w_{L_\tau}, w_q) = (1, 100, 10, 10, 10, 0)$. v2 derives them from a balance principle: given cosmology-specific deposit/drain coefficients per charge, the *symmetric* weights are exactly those for which the per-charge weighted asymmetry vanishes.

Definition 15 (Balance). For *BalanceCoeffs* $b = (\alpha, \omega) : \text{ChargeLabel} \rightarrow \mathbb{Z}$ representing mean-deposit and mean-drain rates, the weight w is *balanced* iff $\forall c. w(c) \cdot (\alpha(c) - \omega(c)) = 0$.

Definition 16 (Derived weight). *derivedWeight* $(b, \sigma)(c) = \sigma \cdot [\alpha(c) = \omega(c)]$, i.e. scale σ on the balanced charges, zero elsewhere.

Theorem 17 (Derived weights are balanced). *For every b and every σ , *derivedWeight* (b, σ) satisfies the balance constraint.*

Theorem 18 (Nondegeneracy iff some balanced charge exists). *derivedWeight* (b, σ) is *nondegenerate* iff $\sigma \neq 0 \wedge \exists c. \alpha(c) = \omega(c)$.

Corollary 19 (Symmetric existence). *A nontrivial balanced weight exists for b iff at least one charge satisfies $\alpha(c) = \omega(c)$.*

The corollary has a non-obvious consequence: in a cosmology where every charge has unequal α and ω rates, the only balanced weight is identically zero, and the game collapses to a 50/50 no-policy-gradient regime. The framework can detect this automatically and flag the cosmology as *degenerate*.

4.5. Physics-extension layer (Mathlib + PhysLean)

The v1.0 “no Mathlib in the core” purity is preserved for the game kernel (Appendix A.1; the audit guarantee survives). The *physics extension*, namespaced `GOG.Physics`, is permitted Mathlib imports and follows PhysLean’s physical-quantity hierarchy [4] and Tooby-Smith’s index-notation extension [5] for the tensor algebra. In particular:

- `Mathlib.MeasureTheory.Integral.Bochner` for Noether currents as Bochner-integrable sections of the appropriate bundle;
- `Mathlib.Geometry.Manifold.SmoothManifoldWithConers` for the $\mathbb{S}^3$ structure and $\text{SO}(4)$ -equivariance lemmas;
- `Mathlib.Analysis.NormedSpace.LpSpace` for the $L^2(\mathbb{S}^3)$ embeddings of the FEEC spaces;
- the PhysLean `LorentzTensor` and `StandardModel` hierarchies for the gauge-sector observables.

The Lean upstream commit pins for these imports are listed in Appendix C.

5. Reward-definability tier system (W-4)

The W-4 conjecture from the v1.1 supplement is structural rather than theorem-shaped: the reward signal is computable at five distinct *rigor tiers*, and league-play comparisons across tiers are an empirical falsifier on the role of axiomatization in policy formation. Table 1 summarizes the tiers; the dissertation deliverable is the cross-tier policy-similarity matrix per move class, with prediction (W-4 hypothesis) that Tier R5 and Tier R1 A-agents agree on at least 80% of move classes by move-selection frequency.

6. The world model: sketch-DiT lifted to $\mathbb{S}^3$

The agent’s world model is a sketch-guided diffusion transformer in the architectural lineage of Camburn 2025 [1], lifted from 2D Cartesian patches to

Tab. 1. Reward-definability tiers (W-4). Each tier is a stricter standard for the rigor of an A/Ω move’s reward signal. Lower R-number is stricter. League-play is tier-stratified; cross-tier comparison matrix is the W-4 deliverable.

Tier	Reward source		Rigor
R5	lattice	transfer-matrix spectrum	combinatorial
R4	lattice	order parameter (Wilson, Polyakov)	numerical
R3	TQFT invariant (Donaldson, S–W, C–S)		topological
R2	AdS-dual classical-gravity quantity		classical GR
R1	continuum AQFT spectral condition		axiomatized

$\text{SE}(4)$ -equivariant patches on simplicial $\mathbb{S}^3$. Camburn’s original architecture treats physics simulation as conditional generation: boundary conditions $\mathcal{B} = \{\text{sketch}, \text{edges}\}$ encode geometry, a ViT-style network with three innovations (spatial-relationship encoder S_{ij} , multi-scale neighborhood attention with $k \in \{1, 2, 4\}$, cross-attention boundary injection at $2,304 = 64 \cdot 3 \cdot 12$ sites) denoises a latent $z \in \mathbb{R}^{1024 \times 32 \times 32}$ via 25-step DDIM with classifier-free guidance scale $w=2.5$, and a VAE decoder produces the steady-state field $F \in \mathbb{R}^{256 \times 256}$. On 1,000 FDTD electromagnetic test cases, mean SSIM = 0.834 ± 0.109 , peak SNR = 25.0 ± 4.2 dB, boundary accuracy 91.1% [1, Tab. 1].

The lift to $\mathbb{S}^3$ proceeds in five steps. (i) *Patches*. The 64-patch grid becomes a partition of the 600-cell-refined triangulation into 64 macro-patches of approximately equal volume; at $n=2$ each patch holds ≈ 600 tetrahedra. (ii) *Spatial-relationship encoder*. The pairwise distance $d_{ij} = \|p_i - p_j\|_2$ becomes the chord distance in ambient $\mathbb{R}^4$; the direction u_{ij} is the unit tangent to the $\mathbb{S}^3$ -geodesic from p_i to p_j . The encoder S_{ij} retains its form; only the metric changes. (iii) *Multi-scale neighborhood attention*. The L^∞ k -neighborhoods become geodesic-ball k -neighborhoods on $\mathbb{S}^3$; locality is well-defined (every point has a normal neighborhood of radius up to $\pi/2$). (iv) *Cross-attention boundary injection*. The $2,304 = 64 \cdot 3 \cdot 12$ injection points become $3,072 = 64 \cdot 4 \cdot 12$ with a fourth boundary-condition channel encoding the metric perturbation $h_{\mu\nu}$. (v) *Equivariance*. The VAE encoder/decoder is replaced by an $\text{SE}(4)$ -equivariant variant via `e3nn` [26]: each feature channel carries a fixed irrep label (ℓ, p) where ℓ is spin and p parity,

and all operations are equivariant by construction.

Loss. Following Camburn’s adaptive-blending strategy $\alpha = 1 - n/1000$ with the addition of a Lean-verification term:

$$\begin{aligned} \mathcal{L}_{\text{total}} = & \mathcal{L}_{\text{diff}} + 0.3 \mathcal{L}_{\text{recon}} + 0.1 \mathcal{L}_{\text{edge}} \\ & + 0.3 \mathcal{L}_{\text{LPIPS}} + 0.4 \mathcal{L}_{\text{prior}} + 0.5 \mathcal{L}_{\text{constr}}. \end{aligned} \quad (7)$$

$\mathcal{L}_{\text{constr}}$ is computed by running the Lean `apply` oracle on a sample of predicted next-states and counting the `none` fraction. This is the formal-verification training signal: the world model is rewarded for predicting physically legal transitions, not merely plausible-looking ones.

Cost. Per-tick world-model rollout on a single H100 is ≈ 80 ms at $n = 2$; the full A/Ω inner loop (observe $\rightarrow$ propose $\rightarrow$ rollout $\rightarrow$ Lean check $\rightarrow$ score) runs at ≈ 2 Hz on a single node and an estimated ≈ 400 Hz on a 256-node IMPALA + Reverb cluster.

7. Mock exhibition match

We exhibit a four-phase annotated match between a frozen A and a frozen Ω on a fixed cosmology. The match is not a trained-agent run; it is a scripted demonstration showing the move grammar, the Lean oracle, the Noether-charge accounting, and the verdict computation in their full form.

Cosmology Ω_0 . Round $\mathbb{S}^3$ at $n = 2$; $\Lambda = 0$ (Minkowski background); SM parameters at the M_Z scale ($g_1 = 0.357$, $g_2 = 0.652$, $g_3 = 1.221$, $v = 246$ GeV); axion dark sector with $m_a = 10 \mu\text{eV}$; low-entropy past hypothesis. Horizon $T = 10$ ticks; threshold $\Theta = 100$; default weights.

The full move log appears in table 2; the verdict matrix (this cosmology vs. a $\Lambda > 0$ counterfactual) appears in table 3.

8. Sharpening four Clay-class problems

GOG does not solve the open problems of mathematical physics. It forces them into precise computable form. table 4 summarizes the four Clay-class problems on which GOG bears.

9. Unified-field substrate (Ghose 2025)

The unified-field sector adopts the helicity-resolved Poisson-process formulation of Ghose [3]. On a spin network Γ embedded in $\mathbb{S}^3$, each directed edge e carries two complex amplitudes $\Psi_e^\pm(t)$. The dynamics is the Gaveau–Jacobson–Kac–Schulman master equation generalized to spin networks ([3, Eq. 27,

28]):

$$\frac{d}{dt} \Psi_e^+ = -\lambda(\Psi_e^+ - \Psi_e^-) + T^+(e), \quad (8)$$

$$\frac{d}{dt} \Psi_e^- = -\lambda(\Psi_e^- - \Psi_e^+) + T^-(e). \quad (9)$$

Under analytic continuation $t \mapsto it$ and introduction of a fundamental length $\ell \sim \ell_P$, this reduces to a Dirac equation in the Weyl representation; the Riemann–Silberstein vector $F = E + iB$ and its non-Abelian generalization extend the construction to the full $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ gauge sector.

Move space on the substrate. A and Ω moves perturb λ and $T^\pm(e)$, nothing more. An A move of class *vacuum-stabilization* increases λ locally, driving the local stochastic dynamics toward the helicity-balanced equilibrium that satisfies the Wheeler–DeWitt-like constraint $\hat{H}\Psi = \sum_e (I - F_e)\Psi_e = 0$ [3, Eq. 33]. An Ω move of class *decoherence* increases the imbalance $|\Psi^+ - \Psi^-|$ by injecting asymmetric transport.

Open. Ghose’s helicity-resolved spin-foam path-integral amplitude ([3, Eq. 34]) is stated formally but not proven well-defined as a sum over histories. We treat the GOG spin-network substrate as the discrete master-equation form (Eqs. 9), which is well-defined; the spin-foam continuum-limit version is in the same status as Ghose puts it, namely a forthcoming-paper construction.

10. Empirical falsifier

Any framework that treats the universe as a discretized computation must answer: does the discretization leave observable traces? Beane, Davoudi & Savage [2] settle the form under the simplest hypothesis (unimproved Wilson-fermion simulation on a hyper-cubic spacetime grid) and extract the bound $b^{-1} \gtrsim 10^{11}$ GeV from the GZK cutoff. The bosonic Wilson dispersion ([2, Eq. 16])

$$\sinh^2(bE_b/2) - \sum_j \sin^2(bk_j/2) - (bm_b/2)^2 = 0 \quad (10)$$

has only cubic, not full rotational, symmetry. The Δ -resonance interaction rate between a UHE proton and a CMB photon picks up an angular-dependent correction:

$$\omega \sim \omega_0 \left[1 + \frac{\sqrt{\pi} b^2 |p|^2}{9} \left(Y_4^0 + \sqrt{\frac{5}{14}} (Y_4^{+4} + Y_4^{-4}) \right) \right] + \dots \quad (11)$$

The $(Y_4^0, Y_4^{\pm 4})$ combination is the unique cubic-invariant $\ell=4$ angular pattern.

Tab. 2. Mock exhibition match: move log. Frozen A vs frozen Ω on Ω_0 (round $\mathbb{S}^3$, $n=2$, $\Lambda=0$). Reward column is per-tick contribution to R ; cumulative shown in last column.

#	by	class	action	ΔQ	r_t	$\sum r$
1	A	stellar aggregation	trigger Jeans collapse, $M \gtrsim 10^{29}$ kg	$\Delta E \approx +1$	+1.0	+1.0
2	Ω	turbulence injection	Kolmogorov cascade at $L=10^6$ m	$\Delta E \approx -0.3$	-0.3	+0.7
3	A	controlled hadronization	guide QGP through QCD crossover, $\Delta B=+3$	$\Delta B=+3$	+300	+300.7
4	Ω	EW reheat	local heating to $T > 160$ GeV	$\Delta E = -0.5$	-0.5	+300.2
5	Ω	sphaleron drive	$B+L$ violation, convert 2 baryons $\rightarrow$ 2 e-leptons	$\Delta B = -2, \Delta L_e = +2$	-180	+120.2
6	Ω	EW quench	rapid cool below T_{EW}	$\Delta E \approx 0$	0	+120.2
7	A	vacuum stabilization	+10% to λ_H in high-asset region	$\Delta E = +0.5$	+0.5	+120.7
8	A	Skyrmion deposit	restore $\Delta B = +2$	$\Delta B = +2$	+200	+320.7
9	A	gauge binding	4 nucleons $\rightarrow$ He-4, $\Delta E = 28.3$ MeV	$\Delta E = +0.3$	+0.3	+321.0
10	—	horizon	verdict computed	—	—	+321.0

Tab. 3. Verdict matrix. The same move log replayed on two cosmologies. The $\Lambda > 0$ counterfactual shows diminishing returns from late-time structure formation (Hubble friction); Ω 's sphaleron drive drains baryons faster than A replaces them.

	Ω_0 ($\Lambda=0$)	Ω'_0 ($\Lambda>0$)
$\sum R$	321	78
constraints hold	true	true
final B	3	0
α Wins	true	false
ω Wins	false	true (cl. 1,2)
classification	habitable	Ω -favored

GOG commitment. A GOG-trained model that claims to reproduce our universe must produce a UHECR sky distribution compatible with Auger / Telescope-Array isotropy bounds. The 600-cell-refined arena has icosahedral H_4 symmetry, not cubic; its associated angular signatures are $\ell=12$ and $\ell=30$ (the lowest non-trivial H_4 irreps). The Beane–Davoudi–Savage bound therefore does not apply directly; instead, the GOG-specific bound is on the H_4 -invariant $\ell=12$ amplitude—an analysis not yet performed in the literature, which we name as a deliverable.

11. Roadmap, costs, named open dependencies

11.1. Named open conjectures

1. Conjecture C-26 (this work): five Sakharov branches $\rightarrow$ five basins in trained-A policy space. *Empirical, settled by months 48–60.*
2. FEEC–Yang–Mills mass-gap convergence (this work): Δ_h converges to a finite limit ≥ 1.5 GeV

as $h \rightarrow 0$. *Game-mechanical version is now Lean theorem schema (v2); continuum-limit convergence remains empirical.*

3. Yang–Mills mass gap (Clay). *Untouched, no near-term expectation.*
4. Smooth Poincaré 4D. *Activated only by topology-changing moves; deferred.*
5. Asymptotic-safety completion of GR. *Active research.*
6. Wightman / AQFT axiomatization on globally hyperbolic Lorentzian manifolds. *Active research; not GOG-blocking.*
7. Loop-refined 600-cell isoperimetric constants $h(G_n)$ (NEW v2): graph-combinatorics computation, Phase-1 deliverable.
8. Joint substrate consistency under A/ Ω axiom moves (NEW v2): out of scope for v1.1, named for v1.2.

A. Lean 4 source

A.1. v1.0 game-theoretic core (GOG.Core)

The full v1.0 source is reproduced in [29] and the prior submission; the central definitions are recapitulated in §3. The file compiles against Lean 4 v4.18.0 with no Mathlib dependency. `#print axioms` output (excerpted):

```
'GOG.noetherReward_telescopes' : [propxt, Quot.sound]
'GOG.noetherReward_charges_only' : (no axioms)
'GOG.accReward_eq_endpoints' : [propxt, Quot.sound]
'GOG.game_is_complete' : [propxt, Classical.choice,
  Quot.sound]
'GOG.alpha_deposit_reward' : [propxt, Quot.sound]
'GOG.vacuum_self_reward_zero' : [propxt]
```

Tab. 4. Clay-class problems sharpened by GOG. Each row exhibits the (i) Clay statement, (ii) the discrete GOG-version that v2 has formalized, and (iii) the falsification mode.

Problem	Clay statement (continuum)	GOG discrete version	Falsification
Yang–Mills mass gap	prove $\exists \Delta > 0$: $\text{spec}(H) \setminus \{0\} \geq \Delta$ on $\mathbb{R}^4$ for compact simple G	<i>GapStab/AlphaCert</i> schema, §4.1–§4.2; instantiable per refinement level	$h(G_n) \rightarrow 0$ across some refinement sequence
Navier–Stokes regularity	global existence + smoothness for incompressible NS in 3D	FEEC-discrete NS solver maintains bounded vorticity from smooth ID $\forall t$	blow-up persisting across mesh refinements
Hodge conjecture	every Hodge class is $\mathbb{Q}$ -linear combination of algebraic cycles	trivial for $\mathbb{S}^3$; deferred to non-trivial Thurston geometries (Nil, Sol, $\mathbb{H}^2 \times \mathbb{R}$)	not GOG-blocking on default arena
Smooth Poincaré 4D	every smooth $M^4 \cong_{\text{TOP}} \mathbb{S}^4$ is diffeomorphic to $\mathbb{S}^4$	load-bearing only when topology-changing moves activate (Tier-2+)	discrete surgery on triangulation gives non-equivalent smooth structure

Tab. 5. 60-month phased plan (v2 revised estimates). Phases inherit Pass-3 scope; v2 adjustments are listed in the deliverables column. Total: ≈ 100 FTE-yr; $\approx 3 \times 10^{23}$ FLOPs.

Months	Phase	Deliverables	FTE-yr / FLOPs
0–6	infrastructure	Lean v1.0 core (extant); v1.1 <i>GapShape/Cheeger/MetaAxiom/Weights</i> (this draft); FEEC mesh manager on 600-cell; CCZ4 integrator; $h(G_0)$, $h(G_1)$ exhaustive Cheeger constants	8 / negligible
6–18	discrete prototype (T5–T4)	MJX backend; SE(4)-equivariant agent network; sketch-DiT lift on $\mathbb{S}^3$ at $n=1$; reward-tier R5 league	12 / 3×10^{21}
18–36	Tier-3 buildout	domain-wall fermions on FEEC- $\mathbb{S}^3$; Lefschetz-thimble solver; full sketch-DiT at $n=2$; reward-tier R3 cross-comparison	30 / 8×10^{22}
36–48	Tier-2 + open-endedness	POET-style Ω_0 -mutation co-evolution; instanton/-monopole solitons; hierarchical RL across cosmologies	20 / 1.5×10^{23}
48–60	Tier-1 + Conjecture C-26	Sakharov-branch curriculum; A league with diverse seeds; cluster trained agents in policy-network feature space; check basin count $\stackrel{?}{=} 5$	30 / 8×10^{22}

A.2. v1.1 extensions (GOG.CoreV11)

The v1.1 file (381 lines, ≈ 14 KB) compiles against Lean 4 v4.18.0 with no Mathlib dependency, no sorry placeholders, and no custom axioms beyond the stock three. The four modules (*GapShape*, *Cheeger*, *MetaAxiom*, *Weights*) are presented in §4. We reproduce a representative excerpt (the *GapShape* deposit construction):

```
def GapStab.deposit (g : GapStab) (n : Level) (k : Nat) : GapStab :=
{ baseline := fun m =>
  Nat.max (g.baseline m) (if n <= m then k else 0)
, monotone := by
  intro m
  have hg : g.baseline m <= g.baseline (m + 1) := g.monotone m
  have hk : (if n <= m then k else 0) <= (if n <= m + 1 then k else 0) := by
    by_cases hnm : n <= m
    . have hnm1 : n <= m + 1 := Nat.le_succ_of_le hnm
```

```
simp only [hnm, hnm1, if_true]
exact Nat.le_refl k
. by_cases hnm1 : n <= m + 1
. simp only [hnm, if_false, hnm1, if_true]
  exact Nat.zero_le k
. simp only [hnm, if_false, hnm1, if_false]
  exact Nat.le_refl 0
apply Nat.max_le.mpr
refine <., .>
. exact Nat.le_trans hg (Nat.le_max_left _ _)
. exact Nat.le_trans hk (Nat.le_max_right _ _) }
```

The full source is published as a companion artifact (GOG_CoreV11.lean); the verbatim #print axioms output is in Appendix B.

B. #print axioms (verbatim)

```
GOG.GapShape.deposit_dominates : [propept]
GOG.GapShape.deposit_raises_at_or_above : [propept]
GOG.GapShape.deposit_compose_monotone : [propept]
GOG.Cheeger.cert_yields_deposit : [propept]
GOG.Cheeger.cert_carries_up : [propept]
```

```

GOG.MetaAxiom.substrateCostAfter_monotone : [propext]
GOG.MetaAxiom.substrateCostAfter_le_append_left : [
  propext]
GOG.MetaAxiom.alphaMaxCost_le_substrate : [propext]
GOG.MetaAxiom.omegaMaxCost_le_substrate : [propext]
GOG.MetaAxiom.substrateCost_eq_max : [propext]
GOG.Weights.zero_balanced : [propext]
GOG.Weights.derivedWeight_balanced : [propext, Quot.
  sound]
GOG.Weights.derivedWeight_nondeg_iff : [propext]
GOG.Weights.symmetric_exists_iff : [propext, Quot.
  sound]

```

12 of 14 v1.1 theorems use only `propext`; two use `propext+Quot.sound`. None use `Classical.choice`—an improvement over v1.0’s `game_is_complete`, which used `by_cases`. All three remain Lean’s foundational axioms; no GOG-specific axioms have been introduced.

C. Reproducibility

Software stack. Lean 4 toolchain v4.18.0, pinned in `lean-toolchain`. *Game core*: no Mathlib. *Physics extension*: Mathlib4 (commit pinned in `lakefile.lean`); PhysLean v0.4 [4]; $e3nn \geq 0.5$. $JAX \geq 0.4$ with CUDA 12.x; PyTorch ≥ 2.4 ; MJX (MuJoCo XLA). Einstein Toolkit [27] and openQCD for differential CI. IMPALA + Reverb + Ray for distributed RL. All floating-point uses IEEE 754 round-to-nearest-even; deterministic CUDA kernels only.

Replay protocol. Every A/Ω game emits a replay log containing (i) tournament parameters (w, Θ, T) ; (ii) cosmology Ω_0 (19 SM parameters, Λ , initial 3-metric, dark sector, low-entropy past flag); (iii) PRNG seed; (iv) full move log (one line per move: tick, player, label, parameters, Lean `apply` result). Verdict computation from the replay log is deterministic and bit-identical across reruns of the same hardware/software stack.

Hardware. Tier-3 training run ($\approx 8 \times 10^{22}$ FLOPs) fits on a 256-node H100 cluster running ≈ 30 days at 80% utilization. The Lean v1.1 core compilation runs on a single CPU in ≈ 2 s; the full PDF builds in ≈ 30 s on a single CPU.

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